SUPERSTORE SALES PERFORMANCE ANALYSIS USING SQL

Project Description

Analyzed 9,694 retail sales records using SQL to study customer behavior, sales trends, product performance, and profitability insights. Performed data extraction, filtering, and aggregation to generate meaningful business insights and support data-driven decision-making.

Objectives

- Analyze overall sales performance
- Identify top-performing products and categories
- Study customer purchasing behavior
- Analyze regional performance
- Find factors affecting profit

Dataset Information

- Dataset: Sample Superstore Dataset
- Total Records: 9,694
- Tool Used: MySQL Workbench
- Database: superstore_project
- Table Name: superstore_dataset

Query Log

Creating the database

- `CREATE DATABASE superstore_project;`

Using the created database superstore_project

- `USE superstore_project;`

Query 1: Display First 10 Records of the Dataset

Purpose: To verify whether the dataset was imported successfully and examine the data structure.

```
SELECT * FROM superstore_dataset LIMIT 10;
```

Observation: The first 10 records were displayed successfully, confirming that the dataset was imported correctly.

Explanation: I used LIMIT 10 to inspect a sample of the dataset before starting the analysis. This helps verify column names and data quality.

Query 2: Total Number of Records in Dataset

Purpose: To determine the total number of records available in the dataset.

```
SELECT COUNT(*) AS total_rows FROM superstore_dataset;
```

Output: 9694

Observation: The dataset contains 9,694 records.

Explanation: I used COUNT(*) to understand the dataset size before performing further analysis.

Query 3: View Dataset Structure

Purpose: To examine column names and data types.

```
DESCRIBE superstore_dataset;
```

Observation: The dataset structure, column names, and data types were displayed successfully.

Explanation: I used DESCRIBE to understand the schema and determine which columns would be used for analysis.

Query 4: Calculate Total Sales

Purpose: To calculate the total sales value from the superstore_dataset.

```
SELECT ROUND(SUM(Sales),2) AS total_sales FROM superstore_dataset;
```

Observation: The total sales value was calculated successfully using the SUM() function.

Explanation: I used SUM() to calculate the overall sales value from the dataset and ROUND() to limit the output to two decimal places improving readability and making the result more suitable for reports and business analysis.

Query 5: Calculate Total Profit

Purpose: To calculate the overall profit earned by the superstore.

```
SELECT ROUND(SUM(Profit),2) AS total_profit FROM superstore_dataset;
```

Output: 282857.75

Observation: The superstore earned a total profit of **282857.75** from all sales transactions recorded in the dataset.

Explanation: I used the SUM() function to add all profit values from the dataset and ROUND() to display the result up to two decimal places. This query helps measure the overall profitability of the business.

Query 6: Sales by Region

Purpose: To identify which region generated the highest sales.

```
SELECT Region, ROUND(SUM(Sales),2) AS total_sales FROM superstore_dataset GROUP
BY Region ORDER BY total_sales DESC;
```

Output:

Region	Total Sales
West	713471.34
East	672194.05
Central	497800.87
South	388983.59

Observation: The **West** region generated the highest sales of **713471.34**, followed by the **East** region with **672194.05**. The **South** region recorded the lowest sales of **388983.59**.

Explanation: I used the GROUP BY clause to group records according to regions and the SUM() function to calculate total sales for each region. The ORDER BY clause was used to display the regions in descending order of sales, making it easier to identify the best-performing region.

Query 7: Sales by Category

Purpose: To identify the category with the highest sales.

```
SELECT Category, ROUND(SUM(Sales),2) AS total_sales FROM superstore_dataset
GROUP BY Category ORDER BY total_sales DESC;
```

Output:

Category	Total Sales
Technology	835900.07
Furniture	733046.86
Office Supplies	703502.93

Observation: Technology recorded the highest sales, while Office Supplies recorded the lowest sales.

Explanation: SUM() was used to calculate total sales for each category and GROUP BY was used to group the data by category.

Query 8: Profit by Region

Purpose: To analyze profit generated by each region.

```
SELECT Region, ROUND(SUM(Profit),2) AS total_profit FROM superstore_dataset
GROUP BY Region ORDER BY total_profit DESC;
```

Output:

Region	Total Profit
West	106021.15

Region	Total Profit
East	90672.01
South	46035.69
Central	40128.90

Observation: West generated the highest profit, while Central generated the lowest profit.

Explanation: SUM() was used to calculate total profit and GROUP BY was used to group the data by region.

Query 9: Profit by Category

Purpose: To identify which category generated the highest profit.

```
SELECT Category, ROUND(SUM(Profit),2) AS total_profit FROM superstore_dataset
GROUP BY Category ORDER BY total_profit DESC;
```

Output:

Category	Total Profit
Technology	145387.10
Office Supplies	120489.89
Furniture	16980.77

Observation: Technology generated the highest profit, while Furniture generated the lowest profit.

Explanation: SUM() was used to calculate total profit and GROUP BY was used to group the data by category.

Query 10: Top 10 Best-Selling Products

Purpose: To identify the products with the highest sales.

```
SELECT `Product Name`, ROUND(SUM(Sales),2) AS Total_Sales FROM
superstore_dataset GROUP BY `Product Name` ORDER BY Total_Sales DESC LIMIT 10;
```

Output: Top 10 products were displayed based on total sales.

Observation: Canon imageCLASS 2200 Advanced Copier recorded the highest sales of **61599.82** among all products.

Explanation: SUM() was used to calculate total sales for each product, and ORDER BY was used to display the products with the highest sales first.

Query 11: Top 10 Most Profitable Products

Purpose: To identify the products that generated the highest profit.

```
SELECT `Product Name`,ROUND(SUM(Profit),2) AS Total_Profit FROM superstore_dataset GROUP BY `Product Name` ORDER BY Total_Profit DESC LIMIT 10;
```

Output: Top 10 products were displayed based on total profit.

Observation: Canon imageCLASS 2200 Advanced Copier generated the highest profit of **25199.93**.

Explanation: SUM() was used to calculate total profit for each product and ORDER BY was used to display the most profitable products first.

Query 12: Top Customers by Sales

Purpose: To identify customers who generated the highest sales.

```
SELECT `Customer Name`, ROUND(SUM(Sales),2) AS Total_Sales FROM superstore_dataset GROUP BY `Customer Name` ORDER BY Total_Sales DESC LIMIT 10;
```

Output: Top 10 customers were displayed based on total sales.

Observation: Sean Miller generated the highest sales of **25043.05** among all customers.

Explanation: SUM() was used to calculate total sales for each customer and ORDER BY was used to display the customers with the highest sales first.

Query 13: Average Sales Per Order

Purpose: To calculate the average sales value per order.

```
SELECT ROUND(AVG(Sales),2) AS average_salesFROM superstore_dataset;
```

Output: 234.42

Observation: The average sales value was calculated successfully.

Explanation: AVG() computes the mean sales value across all records.

Query 14: Sales by Customer Segment

Purpose: To analyze sales across different customer segments.

```
SELECT Segment, ROUND(SUM(Sales),2) AS Total_Sales FROM superstore_dataset GROUP BY Segment ORDER BY Total_Sales DESC;
```

Output:

Segment	Total Sales
Consumer	1150166.18
Corporate	696604.51

Segment	Total Sales
Home Office	425679.16

Observation: The **Consumer** segment generated the highest sales, while the **Home Office** segment generated the lowest sales.

Explanation: SUM() was used to calculate total sales and GROUP BY was used to group the data by customer segment.

Query 15: Top 10 Loss-Making Products

Purpose: To identify products that generated the highest losses.

```
SELECT `Product Name`, ROUND(SUM(Profit),2) AS Total_Profit FROM superstore_dataset GROUP BY `Product Name` ORDER BY Total_Profit ASC LIMIT 10;
```

Output: Top 10 loss-making products were displayed based on total profit.

Observation: **Cubify CubeX 3D Printer Double Head Print** recorded the highest loss of **-8879.97**.

Explanation: SUM() was used to calculate total profit for each product and ORDER BY ASC was used to display the products with the lowest profit first.